

R823-HG-T004

Del modul al classe de representation: §4, parte 1a

Status: provisional construction test. This is a source-keyed translation experiment, not a native-reviewed standard, not an intelligibility result, and not a completed translation of the R823 volume.

Source boundary. German cumulative authority, lines 21117–21146. Authority SHA-256 begins EE8955BCF7A2639; exact slice SHA-256 begins 2D757BFD661CE638. Every translated source segment is aligned in the accompanying clause map. The continuation from line 21148 is outside this tranche.

§ 4. Le relation inter moduls de representation e representationes

1. *Cata modul de representation determina in maniera unic un classe de representation: un modul direct de representation determina un classe direct, e un modul reciproc determina un classe reciproc (in le abbreviationes del fonte: d. D. M. a d. D. K.; r. D. M. a r. D. K.).*

Le construction es le sequente.

Le multiplication del modul direct de representation $\mathfrak{M}$ per un elemento c de $\mathfrak{o}$ defini un homomorfism de $\mathfrak{M}$ a se mesme. Per causa del lege associativ $(cm)\tau = c(m\tau)$, iste homomorfism admitte le elementos de $\mathbb{T}$ como operadores; dunque illo es un *transformation linear* de $\mathfrak{M}$ a se mesme. Isto significa que le association $m \mapsto cm = m'$ ha le proprietate sequente:

$$\text{Si } y_i \mapsto y'_i (= cy_i), \text{ in tal caso etiam } \sum y_i \tau_i \mapsto \sum y'_i \tau_i \left(= c \sum y_i \tau_i \right).$$

Se defini le suma $\mathfrak{T}_1 + \mathfrak{T}_2$ de duo transformationes linear

$$\mathfrak{T}_1 : m \mapsto m' (= c_1 m) \quad \text{e} \quad \mathfrak{T}_2 : m \mapsto m'' (= c_2 m)$$

como le transformation $m \mapsto m' + m'' (= (c_1 + c_2)m)$, e lor productio $\mathfrak{T}_1 \cdot \mathfrak{T}_2$ como le transformation componite $m \mapsto (m'')' = c_1 c_2 m$. In tal caso, le collection del distinct transformationes linear de $\mathfrak{M}$ generate per le elementos de $\mathfrak{o}$ forma un imagine $\mathfrak{D}$ de $\mathfrak{o}$ obtenite per un homomorfism de anels.

Ora se elige un base arbitrari $x_1, \dots, x_n$ de $\mathfrak{M}$. Le transformation linear generate per c es determinate si se cognosce como le elementos del base x_i es transformate, dunque si se cognosce le matrice $(\gamma_{ij}) = C$ del system de equationes

$$x'_j = cx_j = \sum x_i \gamma_{ij}.$$

Per un altere elemento $x = \sum x_j \tau_j$, se ha directmente $x' = \sum x'_j \tau_j$. Iste matrices C corresponde in maniera unic al distinct transformationes linear, nam le matrices e le transformationes determina un le altere. Le matrices forma, como se vide directmente, un anel isomorf a $\mathfrak{D}$; dunque un imagine de $\mathfrak{o}$ obtenite per un homomorfism de anels, e assi un representation de grado n de $\mathfrak{o}$.

Nota editorial de senso

Le unicitate se refere al *classe de representation*, non a un sol representation. In le continuation del fonte, bases differente del mesme modul produce le representationes differente del classe. Le phrase *de $\mathfrak{M}$ a se mesme* declara un application con dominio e codominio $\mathfrak{M}$; illo non adde un affirmation de surjectivitate.

Le productio $\mathfrak{T}_1\mathfrak{T}_2$ conserva le convention del fonte e su resultado explicate c_1c_2m . Nulle convention moderne de composition pote inverter iste ordine silentiose. Le forma test *base* es limitate al base del modul libere in le formulas coordinate; le senso ordinari de fundament non es admittite. *Imagine* nomina le anel resultante, durante que *homomorfism* nomina le application.

Limites del tranche

Le grammatica e le vocabulario es candidatos de construction. Nulle affirmation de comprehension human o de intelligibilitate marginal es facite. Le proxime unitate comencia al linea 21148 e continua le demonstration de §4.